

CSC 221 (D01A)
Introduction to Problem Solving and Programming
Course Syllabus

Instructors	Jeff Elkner and Chris Jones
Session	Fall 2026
Meeting Days	B Days
Times	11:32 am to 12:55 pm
Location	Grace Hopper Center - Rooms 406 and 408
Contact	jde232@email.vccs.edu / cmj2310@email.vccs.edu

Course Description

Introduces problem solving and implementation of solutions using a high level programming language in a structured programming environment. Includes concepts and practice of structured programming, problem-solving, top-down design of algorithms, a high level programming language syntax, control structures, arrays, and an introduction into object oriented programming. First course in a three-course sequence (CSC 221, CSC 222, CSC 223). The assignments in this course require mathematical problem solving skills, algebraic modeling and functions, and use of variables. 3 credits.

General Course Purpose

CSC 221, CSC 222, and CSC 223 comprise the standard sequence of minimal programming content for computer science majors. The course sequence will teach the students to use high-level languages and their applications to problem solving by using algorithms within procedural and object-oriented techniques, while ensuring data adheres to a structured model. This course is the first course in the sequence. It introduces computer based problem solving and implementation of solutions in a high level programming language. Python will be used as the language for this course, and students will sit for the [Certified Entry-Level Python Programmer](#) exam as part of the course.

Course Prerequisites/Co-Requisites

None.

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Course Objectives

- Civic Engagement
 - Engage and build technology that responds to human needs and helps people navigate institutional systems
- Critical Thinking
 - Assess why certain solutions might not work and to save time in coming up with a more efficient approach
- Professional Readiness
 - Work well with others and display situationally and culturally appropriate demeanor and behavior.
- Quantitative Literacy
 - Perform accurate calculations, interpret quantitative information, apply and analyze relevant numerical data, and use results to support conclusions.
- Scientific Literacy
 - Represent real-world objects and processes virtually by identifying properties, behavior, and operations relevant to solving problems on a computer.
- Written Communication
 - Develop, convey, and exchange ideas in writing, as appropriate to a given context and audience.
- Basic concepts of computer systems
 - Differentiate computer components by functionality.
 - Define basics of computer storage devices.
 - Illustrate the computer structure.
 - Define Binary and Hexadecimal numeration systems.
 - Define types of software.
 - Explain the use of computers, and the social impact they have.
 - Discuss secure programming.
 - Evaluate the ethical aspects of programming.
- Processing Code
 - Editors, compilers and/or interpreters; distinguishing source code, object code, and executables
 - Reading and evaluate compilation error messages
 - Executing programs
 - Analyzing and resolving run-time errors

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- Problem analysis and algorithmic modeling
 - List and apply the steps involved in problem solving through algorithmic modeling.
 - Describe activities related to program modeling and design including algorithm development.
 - Solve problems using techniques such as pseudocode, flowcharts, and model development.
 - Verify algorithms and identify errors.
 - Distinguish between procedural techniques and object-oriented techniques.
 - Write programs using good programming practices.
- Use of data
 - Compare and contrast data types.
 - Describe the use of variables.
 - Build expressions using variables, literal data, and operators, correctly using rules of operator precedence.
- Decision structures
 - Describe how conditional selection operations are used to alter the sequential execution of a program.
 - Describe how relational and Boolean operators are used to form logical expressions that evaluate to true or false.
 - Identify techniques to evaluate selection statements for logic errors.
 - Develop programs using sequential and selection operations.
- Repetition structures
 - Describe how repetition structures are used to alter the sequential execution of a program.
 - Choose appropriate repetition structures based on the type of application.
 - Identify techniques to evaluate repetition statements for logic errors.
 - Develop programs using repetition structures.
- Programming with Procedures
 - Apply modularization to manage complexity of programming.
 - Describe the roles of parameters in a procedure definition.
 - Illustrate parameter passing when invoking procedures.
 - Solve problems using procedures.
- Classes and Introduction to Libraries
 - Describe information hiding and encapsulation.
 - Describe the concept of class and object of a class.
 - Use language classes from the standard library to develop programs.

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- Arrays
 - Define the nature and purpose of an array.
 - Use arrays as parameters and returned values in procedures.
 - Evaluate programs that use arrays.
 - Develop applications using arrays.

Major Topics to be Included

- Basic concepts of computer systems
- Processing Code
- Problem analysis and algorithmic modeling
- Use of data
- Decision structures
- Repetition structures
- Programming with Procedures
- Classes and Introduction to Libraries
- Arrays

Required Instructional Materials

- [Beginning Python Programming for Aspiring Web Developers](#) by Jeff Elkner and Chris Jones
- Other freely available resources as provided by instructor

Course Credit: 3 credits

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Policies

I. Expectations

- A. Introduction to Computer Science is a rigorous, college level course that will require sustained and consistent engagement from students.
- B. An average of 90 minutes of homework will be assigned for each 90 minutes in class. We will be utilizing a flipped classroom learning environment, where the lecture portion of the course material will be viewed individually at home *before* class meets, and class time will be used for collaborative engagement and discussion.
- C. Frequent "mini quizzes" at the beginning of class will be used to be sure homework readings and practice have been completed. To be successful in this class, students will be expected to be prepared for these quizzes when they arrive in class.

II. Grading Policies

- A. Grading Scale: A= 100 - 90 B= 89 - 80 C= 79 - 70 D=69 - 60 F= 59 and below
- B. Students will receive a weekly cumulative letter grade that will incorporate daily quizzes, tests, projects, and presentations. These weekly evaluations can be challenged by the student, *but only during the week immediately following when the evaluation is given.*
- C. The average of the weekly evaluations will make up 80% of the final grade, with the course final exam making up 20%.
- D. In cases where district grading policies conflict with college grading policies, the high school and college grades may differ; this may include assignment/test retakes, extended assignment due dates, capped minimum grade allowed, among other such district policies.
- E. It is important that students check their final NOVA grades in SIS as soon as their course is completed.
- F. Course Grade Appeals
 - i. Students who think that a semester grade is in error should contact the instructor immediately to present their concerns. Students who wish to appeal their grade or otherwise report a grievance will need to submit Form 125-021 within 20 days of the end of the semester. The original grade will stand if the student delays in submitting their appeal.
 - ii. NOVA's Student Grievance policy can be found here: https://www.nvcc.edu/policies/_files/608-Student-Grievances.pdf
 - iii. Form 125-021 can be found here: <https://www.nvcc.edu/forms/>

III. Course Policies

- A. Student Rights and Responsibilities
 - i. Students should be familiar with the college's specific expectations concerning the conduct of its students. These expectations apply to all students, including Dual Enrollment students.
 - ii. Student Rights and Responsibilities are outlined in the Student Code of Conduct, found here: <https://www.nvcc.edu/students/handbook/conduct.html>

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B. Academic Integrity

- i. Academic integrity requires that you recognize and acknowledge information derived from others and take credit only for ideas and work that are yours. It should be the guiding principle for all that you do, from taking assessments and making presentations to writing papers. More about academic integrity at NOVA can be reviewed on the Student Conduct and Integrity page on the NOVA website:
<https://www.nvcc.edu/students/handbook/conduct.html>.
- ii. Violating the Academic Integrity Policy will incur consequences. Your instructor may give you a failing grade for the assignment or for the course. Further, you may be referred to NOVA Cares, reported to an academic dean, or even referred to the Dean of Students for disciplinary action depending on how serious an infraction was committed.
- iii. Please review NOVA's Academic Integrity Policy here:
<https://www.nvcc.edu/policies/files/224-Academic-Integrity.pdf>.

C. Accommodations and Accessibility Services

- i. NOVA is committed to ensuring all students have an opportunity to pursue a college education regardless of the presence or absence of a disability. Information on NOVA's Accommodations and Accessibility Services, including how to reach a Accommodations and Accessibility Services counselor, can be found here: <https://www.nvcc.edu/accommodations>.
- ii. Students must reach out to contact NOVA's Accommodations and Accessibility Services to apply for accommodations. A student with a 504 plan or IEP at their high school will still need to apply with NOVA's Accommodations and Accessibility Services – those plans do not automatically carry over to your Dual Enrollment course. If accommodations are agreed upon, students will receive a Memorandum of Accommodation (MOA) by AAS. All information is kept confidential and may increase your chances of success in the academic setting.

D. Advocacy and Privacy of Student Records

- i. Students are expected to reach out to their instructor if they do not understand content or expectations.
- ii. You, as a NOVA student, have a right to review your NOVA grades and other records. Your high school may share grades and other records with NOVA, and NOVA will share your post-secondary (college) grades with your high school. The grades you earn at NOVA are part of a permanent transcript, and you will be required to include your NOVA transcript as part of any future college or graduate school application. NOVA instructors and other college personnel generally may not release a student's educational records without written consent of the student. For dual enrolled students under 18, parents or guardians may generally access records and grades which are created by or shared with a student's high school. For the purposes of these privacy rules, your Dual Enrollment instructor is considered a NOVA employee.
- iii. To grant parents or guardians direct access to NOVA records, students will be required to submit a notarized copy of NOVA Form 125-356, found here:
<https://www.nvcc.edu/forms/pdf/125-356.pdf>.
- iv. For more information about student privacy, parent limitations of access to students' educational records, and other restrictions on sharing students' personally identifiable information, please review NOVA Policy 613 (FERPA):
<https://www.nvcc.edu/policies/files/613-FERPA.pdf>.

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- E. Campus Services
 - i. Dual enrolled students have access to full NOVA campus services to include tutoring, library, and counseling services; student resources are found here:
<http://www.nvcc.edu/students/index.html>.
- F. Office of Wellness and Mental Health
 - i. During your time at NOVA, you may experience challenges including struggles with academics, finances, or your personal well-being. NOVA has support resources available. If you are seeking resources and support or if you are worried about a friend or classmate:
<https://www.nvcc.edu/wellness/index.html>.
- G. Course Drop and Withdrawal Policy
 - i. Please note two important dates related to your enrollment in a course:
 - a. The “Drop” date (also known as census date) for a course is the last day to drop a course. Dropping a course before the drop date will not appear on your NOVA transcript.
 - b. The “Withdrawal” date is the last day to withdraw without a grade penalty. Dropping a course after the drop date and before the withdrawal date will result in a ‘W’ grade appearing on your transcript.
 - c. To identify these dates for your dual enrollment course, please see below on the ‘Course Schedule’ chart or log into your myNOVA account and SIS.
 - ii. Withdrawal Process
 - a. Dual enrolled students are responsible for requesting to drop or withdraw from their DE classes, using Form 125-03, found at the following link:
<https://dashboard.nvcc.edu/Forms/125-03>
 - b. Dual enrolled students will use their myNOVA credentials to access the withdrawal form and will select one or more enrolled DE classes to withdraw.
 - c. The withdrawal form is then routed to the assigned DE instructor and the Office of Dual Enrollment for review and approval.
 - d. Check your VCCS email for the status of your request.
- H. Communication
 - i. Students are required to use their VCCS email accounts (____@email.vccs.edu) to communicate with college personnel and should check their email accounts regularly. Students may access their VCCS email accounts through myNOVA.
- I. Title IX
 - i. Title IX is a civil rights law that prohibits discrimination on the basis of sex in educational programs, activities, admission and employment. Complaints of sex-based discrimination, sexual violence, domestic violence, and sexual or gender-based harassment are governed by the Title IX Policy. For more information about Title IX or to make a report:
<https://www.nvcc.edu/titleix/index.html>.

IV. Additional Course Information

- A. DE students are expected to engage in college level course contents and discussions appropriate for adult learners. Mature topics may be discussed.

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V. Course Schedule

A. Critical Course Dates

Course Start Date	Wednesday, September 2, 2026
Course Drop Date	Tuesday, September 22, 2026
Course Withdrawal Date	Tuesday, November 24, 2026
Final Exam Date	Thursday, December 17, 2026
Course End Date	Wednesday, January 27, 2027

- B. Final Exam/Project Dates:** *The PCEP certification exam will be given on Thursday, December 17, and the final project will be due on the last day of class, Wednesday, January 27.*

NOVA Policies and Procedures

(NOVA Contract Dual Enrollment Syllabus Insert)

Students are responsible for knowing and following College policies, and they should be aware of the [Student Resources](#) available to them. The following are highlights of information that students should be aware of as they begin a course.

Academic Integrity Policy

Accommodations and Accessibility Services

Career Services

Closing Information

Communication

Course Drop/Withdrawal Policy

Enrollment Cancellation

Financial Stability and Advocacy Centers

Forms to Request Services and Assistance

- Accommodations and Accessibility Services: Request for Alternate Format
- Course grade appeal
- Enrollment Services: Course Repeat Request
- Enrollment Services: Enrollment with Permission
- Financial Aid: Satisfactory Academic Progress (SAP) Appeal Form
- Parking Services: Vehicle Registration Form
- Student Grievance Forms
- Veteran Enrollment Request Form (VERF)

Important Dates

Fall 2026 Important Dates

Fall 2026 Breaks & Holidays

Libraries

NOVA All Access

Office of Wellness and Mental Health

Prerequisite Verification Statement

Remote Student Support Services

Student Rights and Responsibilities

Student Integrity and Conduct
Student Ombuds Services (SOS)
TITLE IX
Tutoring

Academic Integrity Policy

NOVA promotes and emphasizes the importance of honesty in academic work. It is therefore imperative for students to maintain the highest standard of honor in their scholastic work.

Academic dishonesty, as outlined in more detail in the [Academic Integrity Policy \(Policy Number: 224\)](#), can include, but is not limited to cheating on an exam or quiz, submitting work that is not your own (plagiarism), or sharing assessments online. Additionally, the use of AI to complete assigned work may be considered a violation of the academic integrity policy. Students are urged to be cautious when using any web tools designed to assist with assignments without their instructors' explicit permission to do so and to cite their sources in all their work.

Consequences of academic dishonesty can include a failing grade on an assignment, a failing grade in the course, and may include additional administrative sanctions such as suspension or expulsion from the college. Procedures for disciplinary measures and appeals are outlined in the [Academic Integrity Procedures](#). It is a student's responsibility to become familiar with the student code of conduct. Lack of awareness is no excuse for noncompliance with NOVA's policies and procedures.

NOVA faculty may choose to use tools such as Turnitin to detect and flag instances of plagiarism in academic writing.

Accommodations and Accessibility Services

NOVA is committed to ensuring all students have an opportunity to pursue a college education regardless of the presence or absence of a disability. No academically qualified student with a disability will be denied access to or participation in the services, programs, and activities of the College. Your access to and inclusion in this course is important to NOVA and your professors. Please request your accommodation letter (Memorandum of Accommodations) early in the semester or as soon as you become registered so that you have adequate time to arrange your approved academic accommodations with your professors. Returning students must renew their Memorandum of Accommodations (MOA) every semester; these students should submit the request 24 hours or later after enrolling in at least one class. Allow up to 7 business days for the request to be approved.

Accommodations are provided for in-person, online, and remote/synchronous (Zoom) learning. To get started, review NOVA's [Accommodation and Accessibility Services](#) website. Following a meeting with a counselor, you will be issued a Memorandum of Accommodation (MOA). You must provide your MOA to your professors, testing proctor, and/or tutoring center to receive your accommodation. You may provide your MOA any time during the semester; however, accommodations are not retroactive. You may email your MOA to your professors or provide them with a printed copy. They will send you an email to acknowledge receipt. You and your professors should schedule a meeting outside of class to discuss questions regarding your MOA. Please remind your professors of any special arrangements that must be made in advance of tests and assessments. If you need a sign language interpreter, or if you need live captions for your Zoom class, send an email to interpreters@nvcc.edu.

Your ability to fully participate in this course matters. If you experience any difficulty accessing course content, activities, or technology, please contact me as soon as possible so that we can address the issue together. Accessibility is a shared responsibility, and your feedback helps ensure that all students have access to learning.

Career Services

The College is committed to providing career services to all students as part of the comprehensive educational journey. [Career Services](#) assists students with exploring, developing and setting goals related to each student's unique educational and academic needs. These services include career assessments, occupational information, goal setting, planning and employment resources. You can [request an appointment with a career counselor](#).

Communication

Northern Virginia Community College (NVCC) faculty, staff, and administrators communicate with students through their official NVCC email accounts (____@nvcc.edu). Students are likewise required to use their VCCS email accounts (____@email.vccs.edu) to communicate with instructors and other college personnel. Students should check their [email accounts](#) regularly.

Course Drop/Withdrawal Policy

Please note these important deadlines related to your enrollment in a course:

- Students may drop courses through [myNOVA](#) until the last day to drop with a tuition refund (census date). Students who drop a class during this period will receive a full refund.
- Students who do not attend at least one class meeting or participate in an online learning class by the last day to drop (census date) may be administratively withdrawn from the class. This means that there will be no record of the class or any letter grade on the student's transcript.
- The *Last Day to Withdraw* is the last day to withdraw from classes without a grade penalty. Students will receive a grade of W. Students may withdraw from a course through the Dual Enrollment Portal. Withdrawing from a course after the census date and before the withdrawal date will result in a "W" grade appearing on your transcript.

To identify important dates for your courses, please visit the [College Academic Calendar](#) and scroll down to the specific session for your course. Please note that any drops or withdrawals from a course may impact [financial aid](#), [international student status](#), or [military benefits](#). Students with questions should check with the appropriate offices.

Financial Stability and Advocacy Centers

[The Financial Stability and Advocacy Center](#) provides assistance to students who are experiencing financial hardships that might prevent the students' academic success. The personnel at the Financial Stability and Advocacy Centers work with students to identify college or community services available. For more information, please visit the [Financial Stability and Advocacy Center](#) webpage, or contact the office by calling 703.323.3450 or emailing financialstability@nvcc.edu.

Forms to Request Services and Assistance

[NOVA's Forms Library](#) is where you can access forms frequently used by the College. The forms are available in Portable Document Format (.pdf), Online or Excel. Some of the forms you will find online include:

- Accommodations and Accessibility Services: Request for Alternate Format
- Course grade appeal
- Enrollment Services: Course Repeat Request
- Enrollment Services: Enrollment with Permission
- Financial Aid: Satisfactory Academic Progress (SAP) Appeal Form
- Parking Services: Vehicle Registration Form
- Student Grievance Forms
- Veteran Enrollment Request Form (VERF)

You also have the option to [Make a Report](#) to the college. Here is the list of reports you can submit:

- Report a [student needed assistance \(non-conduct\)](#).
- Report an [academic integrity violation](#).
- Report a [student conduct violation \(non-academy integrity\)](#).
- Report a [threat assessment](#).
- Report [Sexual Misconduct \(Title IX\)](#).
- [Title IX Pregnant and Parenting Accommodation Request](#).
- Ombuds Assistance Request—student may email OMBUDS@nvcc.edu.

Important Dates

The [academic calendar](#) provides information about important dates and deadlines each semester. Visit this calendar to learn about

- Open registration dates.
- Academic advising dates.
- Registration dates.
- First and Last day of classes
- Deadline to drop the class
- Deadline to withdraw without grade penalty (W)
- Final Exams Schedule.
- Breaks and holidays.
- Deadline to apply for graduation.

* To find the last day to drop (census date) for a course offered in a session go to [schedule of class](#), search for the course, then click on the class number.

**Withdrawals received after the withdrawal deadline require documentation of mitigating circumstances and permission of the instructor and the academic dean. Please check MyNOVA for start dates and census dates for variable duration (dynamic) sessions.

Libraries

[NOVA Libraries](#) support the mission of the College to deliver in-person and online postsecondary teaching, learning, and workforce development to students, faculty, staff and community members. The libraries promote equity and inclusion by addressing systemic barriers to student success and fostering intellectual curiosity, creativity and critical thinking. At our libraries, you may borrow books/printed material, access article databases, access videos, connect to Wi-Fi, use the printers,

and reserve a study room.

Office of Wellness and Mental Health

During your time at NOVA, you may experience challenges including struggles with academics or your personal well-being. NOVA has support resources available. Please contact the [Office of Wellness and Mental Health](#) if you are seeking resources and support, or if you are worried about a friend or classmate.

Prerequisite Verification Statement

As noted in the [Course Prerequisites Policy](#), some courses have prerequisite or corequisite requirements that are established to foster a student's success in the course. Students may not enroll in a course for which they do not meet the prerequisites by the time the course begins or for which they do not simultaneously enroll in any corequisite. Students may be administratively dropped from any course for which they have not met the prerequisite. If a course has a prerequisite, it is the responsibility of the student to ensure completion of this pre-requisite course first. Any student needing assistance in determining prerequisite or corequisite requirements can reach out to their faculty member or Campus Academic Division office for support.

Remote Student Support Services

If you need academic assistance or need college services but cannot make it to campus, many of our services can be accessed remotely. We can provide assistance from several departments, including the following:

- [Academic Advising](#)
- [Canvas and Course Resources](#) (guides and tutorials)
- [Financial Aid](#)
- [IT Help Desk](#)
- [Library Support](#)

Student Rights and Responsibilities

The Office of Student Rights and Responsibilities (OSRR) promotes and encourages honesty, integrity, and respect among NOVA students and staff. We accomplish our mission through education, compliance through behavioral standards, and support of individual rights. Our work helps enhance the quality of the NOVA community and equips students with the knowledge, skills, and resources to be successful inside and outside of the classroom.

The Office of Student Rights and Responsibilities (OSRR) comprise the following areas:

[Student Integrity and Conduct](#)

Investigate alleged violations of the [Student Code of Conduct](#).

- Adjudication of informal and formal Code of Conduct hearings
- Mediation (settling differences)
- Student development workshops
- Professional development workshop on topics like Conflict Resolution and Classroom Management
- Referrals to resources and services

Student Ombuds Services (SOS)

For students who believe an unfair resolution was made for a formal complaint made to NVCC (i.e., academic concerns, ethical behavior, violation of College policies and/or procedures, business services, financial aid, etc.), SOS will investigate the student's concern.

- Confidential - students may discuss problems or issues in a confidential manner
- Independent - of any individual/office within NVCC
- Informal - review matters received and make informal inquiries to offer resolution options, make referrals, and mediate disputes.
- Neutral - strives for impartiality, fairness, and objectivity in the treatment of people and the consideration of issues
- Voluntary – it is the student's responsibility to contact SOS for services

Contact: Dr. Tanneh Kamara

Coordinator, Student Ombuds Services (SOS), Office of Student Rights and Responsibilities

Phone: 703-530-2834

Email: ombuds@nvcc.edu

TITLE IX

Title IX is a civil rights law that prohibits discrimination on the basis of sex in educational programs, activities, admission, and employment. Complaints of sex-based discrimination, sexual violence, domestic violence, dating violence, and sexual or gender-based harassment are governed by the Title IX Policy. For more information or to make a report, visit the [Office of Title IX](#).

Tutoring

NOVA's Tutoring Centers offer free in-person and virtual tutoring to all NOVA Nighthawks. Students can drop by any one of campus Tutoring Centers for walk-in services or set up an appointment for tutoring. To request an appointment for one-on-one tutoring, either in-person or via Zoom, log in to myNOVA to [select EAB Navigate](#). For more information and for Tutoring Center locations, visit [Tutoring & Academic Support](#). In addition, 24/7 online tutoring (via Brainfuse) is free to all students. To access online tutoring, click on the **24/7 Tutoring link** located in your Canvas course in the course navigation menu.